

MCF3M: M5C1

Today:

1. Minds-On: Remember Trigonometry?
2. Finding Side Lengths using SOHCAHTOA
3. Angle Measurements
4. Consolidation: Quiz

Learning Goal:

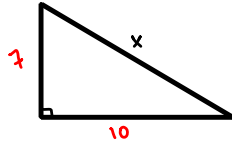
I will use SOHCAHTOA to determine missing sides and angles in right angle triangles

What do you remember about TRIGONOMETRY

Write everything you remember on the padlet
(link in chat)

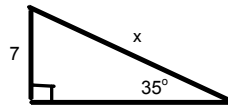


Do we have enough information to determine the missing side of this triangle?



Yes! Use P.T.

What about now?



Yes! Use SOHCAHTOA

Trigonometric ratios allow us to **determine side lengths** using **1 side and 1 angle**

They also allow us to **find missing angles** when we're **given 2 side lengths**

****In right-triangles**

SOH CAH TOA

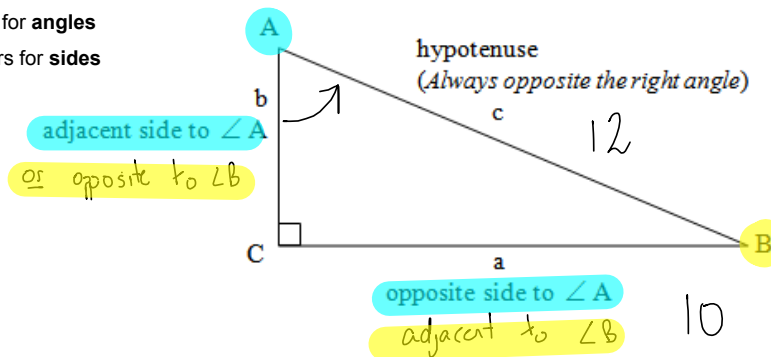
In order to use SOHCAHTOA, we need to be able to label our sides as adjacent, opposite, and hypotenuse. How do we know which is which?

Use the following diagram to aid in identifying a right triangle.

Use **CAPITAL** letters for angles

Use **lower case** letters for sides

$\triangle ABC$



These are the primary trigonometric ratios when we look at $\angle A$:

$$\sin \angle A = \frac{\text{opposite side to } \angle A}{\text{hypotenuse}} = \frac{\text{length of side } a}{\text{length of side } c}$$

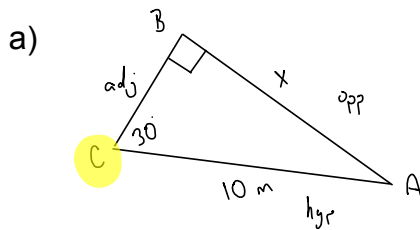
$$\cos \angle A = \frac{\text{adjacent side to } \angle A}{\text{hypotenuse}} = \frac{\text{length of side } b}{\text{length of side } c}$$

$$\tan \angle A = \frac{\text{opposite side to } \angle A}{\text{adjacent side to } \angle A} = \frac{\text{length of side } a}{\text{length of side } b}$$

You Try It: Message Me Your Answer

Your messages will automatically be private and only visible to me

Determine the value of x to 1 decimal place:



$$\sin 30^\circ = \frac{\text{opp}}{\text{hyp}}$$

$$\text{opp} = (\sin 30^\circ) 10$$

$$= 5.0 \text{ m}$$

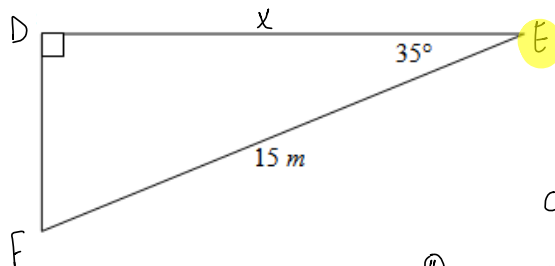
① $\angle C = 30^\circ$
 $b = 10$

② we know $\angle C$

③ SACAHTOA
Use $\sin$

More Practice?

Example: Find the length of x to 1 decimal place.



① we know $\angle E = 35^\circ$
 $d = 15 \text{ m}$

②

③ SACAHTOA

$$\cos E = \frac{\text{adj}}{\text{hyp}}$$

④ $\cos 35^\circ = \frac{\text{adj}}{15}$

⑤ $15 \cos 35^\circ = \text{adj}$
 $\text{adj} = 12.3$

$$\therefore x = 12.3 \text{ m}$$

$$\cos 35^\circ = \frac{a}{15}$$

$$0.819152 = \frac{a}{15}$$

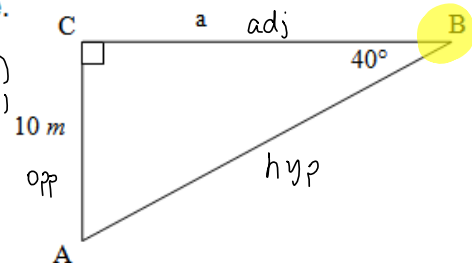
$$12.28728 = a$$

$\therefore$ the length of side a is approximately 12.3 m long.

Example 2: Find the length of side a to 1 decimal place.

SOHCAHTOA

Circle the letters for :- what we know (opp)
- what we want (adj)



$$\tan B = \frac{\text{opp}}{\text{adj}}$$

$$\tan 40^\circ = \frac{10}{a}$$

If the unknown (missing side) is in the **denominator** then you need to use **two steps** to rearrange for the unknown

$$a \tan 40^\circ = 10$$

$$a(\tan 40^\circ) = 10$$

* Evaluate $\tan 40^\circ$ and write as decimal if you'd like
but ALWAYS keep 6 decimal places

$$a = \frac{10}{\tan 40^\circ}$$

$$a = 11.9$$

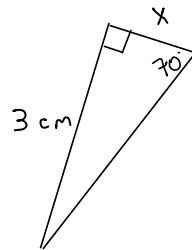
$$\therefore a = 11.9 \text{ m}$$

Since $\tan 40^\circ = 0.8391$,

$$a = 11.91753$$

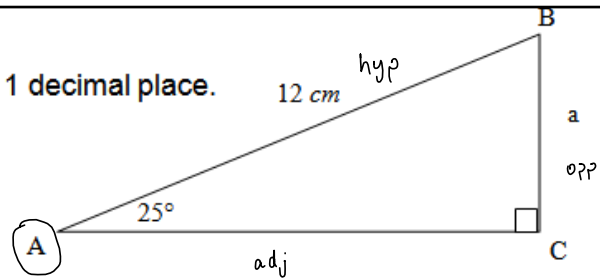
$\therefore$ the length of side a is approximately 11.9 m long.

Determine the value of x :



Example 1: Find the length of side **a** to 1 decimal place.

SOH CAH TOA



- ① • **Identify what we know:** We have $\angle A = 25^\circ$ $c = 12 \text{ cm}$ $\angle C = 90^\circ$
- ② • **Label the triangle:** Label the hypotenuse. Then label triangle in reference to the **angle** we know (A)
- ③ • **Identify correct ratio:** we will need to use **sin**

Circle what we know $\rightarrow$ hyp
 Circle what we want $\rightarrow$ opp (side a)

$$\sin A = \frac{\text{opp}}{\text{hyp}}$$

$$\sin 25^\circ = \frac{\text{opp}}{12}$$

- ④ • Plug in information we know
- ⑤ • Solve for the unknown value:

$$\text{opp} = (\sin 25^\circ \times 12)$$

$$\text{opp} = 5.07$$

$\therefore$ side a is 5.1 cm

$$12(0.422618) = \frac{a}{12} (12)$$

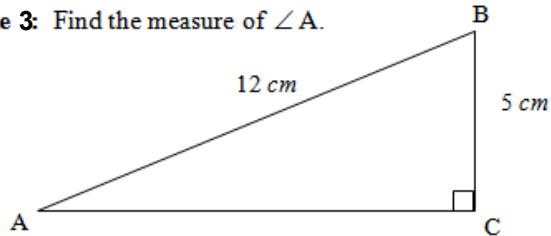
$$5.071416 = a$$

$\therefore$ the length of side a is approximately 5.1 cm long.

Finding Angles in Right Angle Triangles

Example 3: Find the measure of $\angle A$.

(to the nearest degree)



Step 1: Identify what we know:

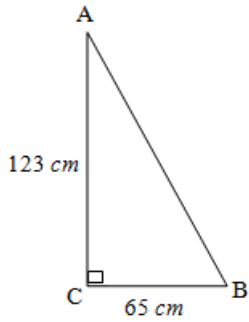
Step 2: Label the Triangle (in reference to either *the angle we know* OR *the angle we want to find*)

Step 3: Identify the correct ratio to use: SOH CAH TOA

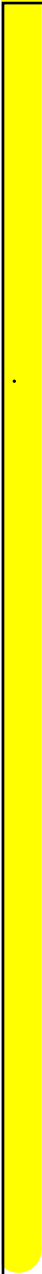
Step 4: Plug in the information we know

Step 5: Solve for the unknown value.

Ex. 4: Solve the following triangle (find all missing side lengths and angle measurements)



Note: "Solve the triangle" means to ...



Consolidation: GoFormative M5C1